

Power series Method

Consider the second order ODE

$$y'' + P(x)y' + Q(x)y = 0 \quad \text{--- (1)}$$

$$\sum_{m=0}^{\infty} a_m (x-x_0)^m$$

→ This is power series centered at x_0 .

- Let $x_0 \in \mathbb{R}$, such that $\lim_{x \rightarrow x_0} P(x)$ and $\lim_{x \rightarrow x_0} Q(x)$ exists.

Then, point x_0 is called ordinary point.

And if x_0 is an ordinary point, then Eqn (1) can have solution of the type $\sum_{m=0}^{\infty} a_m (x-x_0)^m$.

Example:- Use power series method to find the solution of Eqn $(x^2+1)y'' + xy' - xy = 0$ about $x=0$.

Solution $\Rightarrow y'' + \frac{x}{x^2+1}y' - \frac{x}{x^2+1}y = 0 \quad \text{--- (1)}$

$$\lim_{x \rightarrow 0} P(x) = \lim_{x \rightarrow 0} \frac{x}{x^2+1} = 0 \quad Q(x) = \lim_{x \rightarrow 0} \frac{-x}{x^2+1} = 0$$

Let $y = \sum_{m=0}^{\infty} a_m x^m$ be solution of (1)

$$y = a_0 + a_1 x + a_2 x^2 + a_3 x^3 + \dots$$

$$y' = \sum_{m=0}^{\infty} m a_m x^{m-1}$$

$$\rightarrow 0 + a_1 x^0 + 2a_2 x^1 + 3a_3 x^2 + \dots$$

$$y'' = \sum_{m=0}^{\infty} m(m-1) a_m x^{m-2}$$

$$\rightarrow 0 + 0 + 2a_2 x^0 + 3 \times 2 a_3 x + \dots$$

$$(x^2+1)y'' + xy' - xy = 0$$

$$(x^2+1) \sum_{m=0}^{\infty} m(m-1) a_m x^{m-2} + x \sum_{m=0}^{\infty} m a_m x^{m-1} - x \sum_{m=0}^{\infty} a_m x^m = 0$$

$$\dots x^m \dots \sum_{m=0}^{\infty} m(m-1) a_m x^{m-2} + \sum_{m=0}^{\infty} m a_m x^m - \sum_{m=0}^{\infty} a_m x^{m+1} = 0$$

$$\begin{aligned}
 & \sum_{m=0}^{\infty} m(m-1) a_m x^m + \sum_{m=0}^{\infty} m(m-1) a_m x^{m-2} + \sum_{m=0}^{\infty} m a_m x^m - \sum_{m=0}^{\infty} a_m x^{m+1} = 0 \\
 & = \sum_{m=0}^{\infty} m^2 a_m x^m - \sum_{m=0}^{\infty} m a_m x^m + \sum_{m=0}^{\infty} m(m-1) a_m x^{m-2} + \sum_{m=0}^{\infty} m a_m x^m \\
 & \quad - \sum_{m=0}^{\infty} a_m x^{m+1} = 0
 \end{aligned}$$

$$\Rightarrow \sum_{m=0}^{\infty} m^2 a_m x^m + \sum_{m=0}^{\infty} m(m-1) a_m x^{m-2} - \sum_{m=0}^{\infty} a_m x^{m+1} = 0$$

$$\Rightarrow \sum_{m=0}^{\infty} m^2 a_m x^m + \sum_{m=0}^{\infty} (m+2)(m+1) a_{m+2} x^m - \sum_{m=0}^{\infty} a_m x^{m+1} = 0$$

$$\Rightarrow \sum_{m=0}^{\infty} m^2 a_m x^m + \sum_{m=0}^{\infty} (m+2)(m+1) a_{m+2} x^m - \sum_{m=1}^{\infty} a_{m-1} x^m = 0$$

$$\begin{aligned}
 \Rightarrow a_1 x + \sum_{m=2}^{\infty} m^2 a_m x^m + 2a_2 + 6a_3 x + \sum_{m=2}^{\infty} (m+2)(m+1) a_{m+2} x^m \\
 - a_0 x - \sum_{m=2}^{\infty} a_{m-1} x^m = 0
 \end{aligned}$$

$$\Rightarrow 2a_2 + (6a_3 + a_1 - a_0)x + \sum_{m=2}^{\infty} \left[(m+2)(m+1) a_{m+2} + m^2 a_m - a_{m-1} \right] x^m = 0$$

$$\begin{aligned}
 & \Rightarrow a_2 = 0 \\
 & 2a_2 = 0
 \end{aligned}$$

$$6a_3 + a_1 - a_0 = 0 \Rightarrow a_3 = \frac{a_0 - a_1}{6}$$

$$(m+2)(m+1) a_{m+2} + m^2 a_m - a_{m-1} = 0$$

$$a_{m+2} = \frac{a_{m-1} - m^2 a_m}{(m+2)(m+1)}$$

$\boxed{m=2}$

$$a_4 = \frac{a_1 - m^2 a_2}{(m+2)(m+1)} = \frac{1}{12} a_1$$

$$\boxed{m=2}$$

$$a_4 = \frac{a_1 - m^2 a_2}{4 \cdot 3} = \frac{1}{12} a_1$$

$$\boxed{m=3}$$

$$a_5 = \frac{a_2 - 9a_3}{5 \times 4} = \frac{1}{20} \cdot (-1) \left(\frac{a_0 - a_1}{\frac{6}{2}} \right) =$$

Solution

$$y = a_0 + a_1 x + a_2 x^2 + a_3 x^3 + \dots$$

$$y = a_0 + a_1 x + 0 \cdot x^2 + \left(\frac{a_0 - a_1}{6} \right) x^3 + \frac{1}{12} a_1 x^4 + \left(\frac{-3}{40} \right) (a_0 - a_1) x^5 + \dots$$

$$y = \underbrace{a_0 \left(1 + \frac{1}{6} x^3 - \frac{3}{40} x^5 + \dots \right)}_{y_1(x)} + \underbrace{a_1 \left(x - \frac{1}{6} x^3 + \frac{1}{12} x^4 + \frac{3}{40} x^5 + \dots \right)}_{y_2(x)}$$

Example :-

$$y' = 2xy, \quad \text{around } x=0$$

$$\Rightarrow y' - 2xy = 0 \quad \text{--- (1)}$$

$$P(x) = -2x$$

$$\lim_{x \rightarrow 0} P(x) = \lim_{x \rightarrow 0} (-2x) = 0 \Rightarrow x=0 \text{ is an ordinary point.}$$

Let

$$y = \sum_{m=0}^{\infty} a_m x^m \quad \text{be the solution of } y$$

$$y' = \sum_{m=0}^{\infty} m a_m x^{m-1} = \sum_{m=1}^{\infty} m a_m x^{m-1}$$

$$0 = y' - 2xy = \sum_{m=1}^{\infty} m a_m x^{m-1} - 2x \sum_{m=0}^{\infty} a_m x^m = 0$$

$$\sum_{m=1}^{\infty} m a_m x^{m-1} - 2 \sum_{m=0}^{\infty} a_m x^{m+1} = 0$$

$$\sum_{m=1}^{\infty} m a_m x^{m-1} - 2 \sum_{m=0}^{\infty} a_m x^{m+1} = 0$$

$$\Rightarrow \left(\underline{a_1} + 2a_2 x + 3a_3 x^2 + 4a_4 x^3 + \dots \right) - 2 \left(a_0 x + a_1 x^2 + a_2 x^3 + a_3 x^4 + \dots \right) = 0$$

$$\Rightarrow a_1 + \underline{(2a_2 - 2a_0)} x + \underline{(3a_3 - 2a_1)} x^2 + (4a_4 - 2a_2) x^3 + (5a_5 - 2a_3) x^4 + \dots = 0$$

$$a_1 = 0, \quad \begin{cases} 2a_2 - 2a_0 = 0 & , & \underline{3a_3 - 2a_1 = 0} & & 4a_4 - 2a_2 = 0 \\ & & \underline{5a_5 - 2a_3 = 0} & - & - & - \end{cases}$$

$$\textcircled{a_1 = 0}, \quad a_2 = a_0, \quad a_3 = \frac{2}{3} a_1 = 0, \quad 4a_4 = 2a_2 \Rightarrow a_4 = \frac{1}{2} a_2 = \frac{1}{2} a_0$$

$$6a_6 = 2a_4 \\ a_6 = \frac{1}{3} a_4 = \frac{1}{6} a_0$$

$$5a_5 = 2a_3 = 0 \quad a_5 = 0$$

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$$\boxed{(2m+1) a_{2m+1} - 2 a_{2m-1} = 0} \quad \begin{matrix} \rightarrow m=1 \\ \rightarrow m=2 \end{matrix}$$

$$a_{2m+1} = 0 \quad \forall \quad m$$

$$\bullet \quad 2a_2 - 2a_0 = 0$$

$$4a_4 - 2a_2 = 0$$

$$(2m) a_{2m} - 2 a_{2m-2}$$

$$a_{2m} = \frac{2}{2m} a_{2m-2} = \frac{1}{m} a_{2m-2}$$

$$y = a_0 + a_2 x^2 + a_4 x^4 + a_6 x^6 + \dots$$

$$y = a_0 + a_2 x^2 + a_4 x^4 + a_6 x^6 + \dots$$

$$y = a_0 + a_0 x^2 + \frac{a_0}{2} x^4 + \frac{a_0}{6} x^6 + \dots$$

$$y = a_0 \left(1 + x^2 + \frac{x^4}{2} + \frac{x^6}{6} + \dots \right)$$

$$y = a_0 \left(1 + (x^2) + \frac{(x^2)^2}{2!} + \frac{(x^2)^3}{3!} + \dots \right) = a_0 e^{x^2}$$

• Example :- $(1+x^2)y'' + xy' - y = 0$ about $x=0$

$$y'' + \frac{x}{1+x^2} y' - \frac{1}{1+x^2} y = 0$$

$$P(x) = \frac{x}{1+x^2}$$

$$Q(x) = \frac{-1}{1+x^2}$$

$$\lim_{x \rightarrow 0} P(x) = 0$$

$$\lim_{x \rightarrow 0} Q(x) = -1$$

Answer

$$y(x) = a_0 \left(1 + \frac{1}{2} x^2 - \frac{1}{8} x^4 + \frac{1}{16} x^6 - \dots \right) + a_1 x$$

